

## T3 intermediate-EFT one-loop RGE term comparison: EFT\_1 after $F$

This stage has active field content  $\text{SM} + S_1 + S_2$ . The report combines the renormalisable RGBeta running with the dimension-five  $\psi^2\phi^2$  Wilson-coefficient running generated after integrating out  $F$ . Throughout,  $16\pi^2 \mu dX/d\mu = \beta_X^{(1)}$ . For the Wilson sector only the tree-generated stage-1 coefficient  $C^{(0)}$  is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

### Notation key

| symbol                    | interaction / definition                                           |
|---------------------------|--------------------------------------------------------------------|
| $g_1$                     | $D_\mu \supset ig_1 Y B_\mu$                                       |
| $g_2$                     | $D_\mu \supset ig_2 T^A W_\mu^A$                                   |
| $g_3$                     | $D_\mu \supset ig_3 t^A G_\mu^A$                                   |
| $Y_u$                     | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$                           |
| $Y_d$                     | $-\bar{Q} Y_d H d + \text{h.c.}$                                   |
| $Y_e$                     | $-\bar{L} Y_e H e + \text{h.c.}$                                   |
| $y_1$                     | $LF S_1 + \text{h.c.}$                                             |
| $y_2$                     | $LF S_2 + \text{h.c.}$                                             |
| $M_F$                     | $-\bar{F} M_F F$                                                   |
| $T$                       | $\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$             | $\text{Tr}(y_1^\dagger y_1)$                                       |
| $T_\nu^{(2)}$             | $\text{Tr}(y_2^\dagger y_2)$                                       |
| $T_\nu$                   | $T_\nu^{(1)} + T_\nu^{(2)}$                                        |
| $m_1^2$                   | $m_1^2 S_1^\dagger S_1$                                            |
| $m_2^2$                   | $m_2^2 S_2^\dagger S_2$                                            |
| $\lambda_1$               | $\frac{1}{2} \lambda_1 (H^\dagger H)^2$                            |
| $\lambda_{S_1}^{(1)}$     | $\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$              |
| $\lambda_{S_2}^{(1)}$     | $\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$              |
| $\lambda_{HS_1}^{(1)}$    | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$              |
| $\lambda_{HS_2}^{(1)}$    | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$              |
| $\lambda_{HS_1}^{(A)}$    | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$      |
| $\lambda_{HS_2}^{(A)}$    | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$      |
| $\lambda_5$               | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$                      |
| $\lambda_{12}^{(1)}$      | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$            |
| $\lambda_{12}^{(A)}$      | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$    |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$       |
| $\lambda_{S_1}^{(A)}$     | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$   |
| $\lambda_{S_2}^{(A)}$     | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$   |
| $\alpha$                  | $Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$   |

|                        |                                                      |
|------------------------|------------------------------------------------------|
| $C_2(d)$               | $(d^2-1)/4$                                          |
| $T(d)$                 | $d(d^2-1)/12$                                        |
| $C_{LLS_1S_2}^{ij;AB}$ | $L_iL_jS_1^AS_2^B$                                   |
| $C_{ijab}$             | $L_iL_j\phi_a\phi_b$                                 |
| $C_5^{ij}$             | $(L_i^TC L_j)HH$                                     |
| $R_W$                  | $16\pi^2\beta_{C_5}\supset R_W\lambda_5C_{LLS_1S_2}$ |

Renormalisable EFT1 couplings

$\beta_{g_1}^{(1)}$

Quantity:  $g_1$

$16\pi^2\,\mu\frac{dg_1}{d\mu}=\beta_{g_1}^{(1)}$

| Column Representative term structure |                     |
|--------------------------------------|---------------------|
| T1                                   | $\frac{47g_1^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{47g_1^3}{6}$ |

$\beta_{g_2}^{(1)}$   
Quantity:  $g_2$

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-\frac{5g_2^3}{2}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{5g_2^3}{2}$ |

$\beta_{g_3}^{(1)}$   
Quantity:  $g_3$

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-7g_3^3$                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-7g_3^3$ |

$\beta_{Y_u}^{(1)}$

Quantity:  $Y_u$

$$16\pi^2\,\mu\frac{dY_u}{d\mu}=\beta_{Y_u}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                             |
|--------|-----------------------------------------------------------|
| T1     | $-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$ |
| T2     | $+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$   |
| T3     | $+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                       | T2                                                      | T3                                                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_u)_{ij}$ | $+ \mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_u)_{ij}$ |

Quantity:  $Y_u$

Terms 2 of 3

| Column | Representative term structure                                                   |
|--------|---------------------------------------------------------------------------------|
| T1     | $-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$ |
| T2     | $-\frac{9}{4}g_2^2\ (Y_u)_{ij}$                                                 |
| T3     | $-8g_3^2\ (Y_u)_{ij}$                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                              | T2                              | T3                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------------------|---------------------------------|-----------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_u)_{ij} - (Y_u\ (Y_u)^\dagger\ Y_u)_{ij})$ | $-\frac{9}{4}g_2^2\ (Y_u)_{ij}$ | $-8g_3^2\ (Y_u)_{ij}$ |

Quantity:  $Y_u$

Terms 3 of 3

| Column | Representative term structure   |
|--------|---------------------------------|
| T1     | $-\frac{17}{12}g_1^2(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                              |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{17}{12}g_1^2(Y_u)_{ij}$ |

$\beta_{Y_d}^{(1)}$

**Quantity:**  $Y_d$

$$16\pi^2\,\mu\frac{dY_d}{d\mu}=\beta_{Y_d}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                           |
|--------|---------------------------------------------------------|
| T1     | $\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$     |
| T2     | $+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$ |
| T3     | $+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                  | T2                                                      | T3                                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_d)_{ij}$ | $+3\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_d)_{ij}$ | $+3\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_d)_{ij}$ |

Quantity:  $Y_d$

Terms 2 of 3

| Column | Representative term structure                                                               |
|--------|---------------------------------------------------------------------------------------------|
| T1     | $\displaystyle +\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$ |
| T2     | $\displaystyle -\frac{9}{4}g_2^2\ (Y_d)_{ij}$                                               |
| T3     | $\displaystyle -8g_3^2\ (Y_d)_{ij}$                                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                            | T2                              | T3                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------------------------|---------------------------------|-----------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}((Y_d\ (Y_d)^\dagger\ Y_d)_{ij}-(Y_u\ (Y_u)^\dagger\ Y_d)_{ij})$ | $-\frac{9}{4}g_2^2\ (Y_d)_{ij}$ | $-8g_3^2\ (Y_d)_{ij}$ |

Quantity:  $Y_d$

Terms 3 of 3

| Column | Representative term structure  |
|--------|--------------------------------|
| T1     | $-\frac{5}{12}g_1^2(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{5}{12}g_1^2(Y_d)_{ij}$ |

$\beta_{Y_e}^{(1)}$

Quantity:  $Y_e$

$$16\pi^2\,\mu\frac{dY_e}{d\mu}=\beta_{Y_e}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                             |
|--------|-----------------------------------------------------------|
| T1     | $-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$ |
| T2     | $+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$ |
| T3     | $+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                        | T2                                                        | T3                                                     |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-3\,\mathrm{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)\,(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\Big(Y_u\,(Y_u)^\dagger\Big)\,(Y_e)_{ij}$ | $+\mathrm{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)\,(Y_e)_{ij}$ |

Quantity:  $Y_e$

Terms 2 of 2

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$ |
| T2     | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$                         |
| T3     | $-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                 | T2                                         | T3                                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------------|--------------------------------------------|---------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+\frac{3}{2}\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j}$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_1^2\left(Y_e\right)_{i j}$ |

$\beta_{m_1^2}^{(1)}$

Quantity:  $m_1^2$

$16\pi^2\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 2

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-12g_2^2m_1^2$               |
| T2     | $-6g_1^2m_1^2$                |
| T3     | $+2\lambda_{12}^{(1)}m_2^2$   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1              | T2             | T3                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------|----------------|-----------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-12g_2^2m_1^2$ | $-6g_1^2m_1^2$ | $+2\lambda_{12}^{(1)}m_2^2$ |

Quantity:  $m_1^2$

Terms 2 of 2

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$ |

$$\beta_{m_2^2}^{(1)}$$

Quantity:  $m_2^2$

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $6\lambda_{12}^{(1)}m_1^2$    |
| T2     | $+4\lambda_{S2}^{(1)}m_2^2$   |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$ | $d_{S2}$ | $Y_{S2}$ | $d_F$ | $Y_F$          | T1                         | T2                          |
|-------|----------|----------|----------|----------|----------|-------|----------------|----------------------------|-----------------------------|
| T3-D  | -2       | 3        | -1       | 1        | 0        | 2     | $-\frac{1}{2}$ | $6\lambda_{12}^{(1)}m_1^2$ | $+4\lambda_{S2}^{(1)}m_2^2$ |

$\beta_{\lambda_1}^{(1)}$   
Quantity:  $\lambda_1$

$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu}=\beta_{\lambda_1}^{(1)}$

Terms 1 of 5

| Column | Representative term structure                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T2     | $-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$                                                                          |
| T3     | $-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$                                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                        | T2                                                               | T3                                                              |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_1(12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+12\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | $-12\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\,Y_d\,(Y_d)^\dagger\big)$ | $-4\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ |

Quantity:  $\lambda_1$

Terms 2 of 5

| Column | Representative term structure                                                         |
|--------|---------------------------------------------------------------------------------------|
| T1     | $-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$ |
| T2     | $+4\lambda_5\left(\lambda_5\right)^{*}$                                               |
| T3     | $+ \frac{9g_2^4}{4}$                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                    | T2                                      | T3                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------------------------|-----------------------------------------|----------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-12\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}Y_u\left(Y_u\right)^{\dagger}\right)$ | $+4\lambda_5\left(\lambda_5\right)^{*}$ | $+ \frac{9g_2^4}{4}$ |

Terms 3 of 5

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$ |
| T2     | $-9g_2^2\lambda_1$                               |
| T3     | $+\frac{3g_1^4}{4}$                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                               | T2                 | T3                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------|--------------------|---------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$ | $-9g_2^2\lambda_1$ | $+\frac{3g_1^4}{4}$ |

Quantity:  $\lambda_1$

Terms 4 of 5

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+12\lambda_1^2$              |
| T2     | $+6\lambda_{HS_1}^{(1)2}$     |
| T3     | $+3\lambda_{HS_1}^{(A)2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1               | T2                        | T3                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------|---------------------------|---------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+12\lambda_1^2$ | $+6\lambda_{HS_1}^{(1)2}$ | $+3\lambda_{HS_1}^{(A)2}$ |

Quantity:  $\lambda_1$

Terms 5 of 5

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+2\lambda_{HS_2}^{(1) \ 2}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+2\lambda_{HS_2}^{(1) \ 2}$ |

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Quantity:  $\lambda_{S_1}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Terms 1 of 4

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $21g_2^4$                                                               |
| T2     | $+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$ |
| T3     | $+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1        | T2                                                                      | T3                                                     |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------|-------------------------------------------------------------------------|--------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $21g_2^4$ | $+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$ | $+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 2 of 4

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+12g_1^4$                    |
| T2     | $+2\lambda_{12}^{(1)2}$       |
| T3     | $+4\lambda_{HS_1}^{(1)2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1         | T2                      | T3                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------|-------------------------|---------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+12g_1^4$ | $+2\lambda_{12}^{(1)2}$ | $+4\lambda_{HS_1}^{(1)2}$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 3 of 4

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+\frac{3\lambda_{HS_1}^{(A)^2}}{2}$        |
| T2     | $+14\lambda_{S_1}^{(1)^2}$                  |
| T3     | $+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                   | T2                         | T3                                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------|----------------------------|---------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+\frac{3\lambda_{HS_1}^{(A)^2}}{2}$ | $+14\lambda_{S_1}^{(1)^2}$ | $+14\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$ |

Quantity:  $\lambda_{S_1}^{(1)}$

Terms 4 of 4

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+26\lambda_{S_1}^{(A)2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+26\lambda_{S_1}^{(A)2}$ |

$\beta_{\lambda_{S_2}}^{(1)}$

Quantity:  $\lambda_{S_2}^{(1)}$

$$16\pi^2 \mu \frac{d\lambda_{S_2}^{(1)}}{d\mu} = \beta_{\lambda_{S_2}}^{(1)}$$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $6\lambda_{12}^{(1)2}$        |
| T2     | $+4\lambda_{HS_2}^{(1)2}$     |
| T3     | $+10\lambda_{S_2}^{(1)2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                     | T2                        | T3                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------|---------------------------|---------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $6\lambda_{12}^{(1)2}$ | $+4\lambda_{HS_2}^{(1)2}$ | $+10\lambda_{S_2}^{(1)2}$ |

$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$

Quantity:  $\lambda_{HS_1}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$

Terms 1 of 4

| Column | Representative term structure                                                                                                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{HS_1}^{(1)}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{(A)}\right)$ |
| T2     | $+4\lambda_5\left(\lambda_5\right)^*$                                                                                                                                                                            |
| T3     | $+6g_2^4$                                                                                                                                                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                               | T2                                    | T3        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_{HS_1}^{(1)}\left(6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+4\lambda_{S_1}^{(A)}\right)$ | $+4\lambda_5\left(\lambda_5\right)^*$ | $+6g_2^4$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 2 of 4

| Column | Representative term structure            |
|--------|------------------------------------------|
| T1     | $-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$ |
| T2     | $+3g_1^4$                                |
| T3     | $-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                       | T2        | T3                                       |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------|-----------|------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$ | $+3g_1^4$ | $-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 3 of 4

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$ |
| T2     | $+6\lambda_1\lambda_{HS_1}^{(1)}$          |
| T3     | $+4\lambda_{HS_1}^{(1)^2}$                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                         | T2                                | T3                         |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------|-----------------------------------|----------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$ | $+6\lambda_1\lambda_{HS_1}^{(1)}$ | $+4\lambda_{HS_1}^{(1)^2}$ |

Quantity:  $\lambda_{HS_1}^{(1)}$

Terms 4 of 4

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$ |
| T2     | $+6\lambda_{HS_1}^{(A) \ 2}$                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                          | T2                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------|------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{HS_1}^{(A) \ 2}$ |

$\beta^{(1)}_{\lambda_{HS_2}^{(1)}}$

Quantity:  $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta^{(1)}_{\lambda_{HS_2}^{(1)}}$$

Terms 1 of 3

| Column | Representative term structure                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{HS_2}^{(1)}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ |
| T2     | $+12\lambda_5\ (\lambda_5)^*$                                                                                                                |
| T3     | $-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$                                                                                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                           | T2                            | T3                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_{HS_2}^{(1)}(6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $+12\lambda_5\ (\lambda_5)^*$ | $-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$ |

Quantity:  $\lambda_{HS_2}^{(1)}$

Terms 2 of 3

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$    |
| T2     | $+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$ |
| T3     | $+6\lambda_1\lambda_{HS_2}^{(1)}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                      | T2                                         | T3                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------|--------------------------------------------|-----------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$ | $+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$ | $+6\lambda_1\lambda_{HS_2}^{(1)}$ |

Quantity:  $\lambda_{HS_2}^{(1)}$

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $+4\lambda_{HS_2}^{(1) \ 2}$                 |
| T2     | $+4\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                           | T2                                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------|----------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1) \ 2}$ | $+4\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$ |

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity:  $\lambda_{12}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$

Terms 1 of 3

| Column | Representative term structure        |
|--------|--------------------------------------|
| T1     | $4\lambda_5\left(\lambda_5\right)^*$ |
| T2     | $-12g_2^2\lambda_{12}^{(1)}$         |
| T3     | $-6g_1^2\lambda_{12}^{(1)}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                   | T2                           | T3                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------|------------------------------|-----------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $4\lambda_5\left(\lambda_5\right)^*$ | $-12g_2^2\lambda_{12}^{(1)}$ | $-6g_1^2\lambda_{12}^{(1)}$ |

Quantity:  $\lambda_{12}^{(1)}$

Terms 2 of 3

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $+4\lambda_{12}^{(1)2}$                   |
| T2     | $+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ |
| T3     | $+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                      | T2                                        | T3                                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------|-------------------------------------------|-------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+4\lambda_{12}^{(1)2}$ | $+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+4\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$ |

Quantity:  $\lambda_{12}^{(1)}$

Terms 3 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$    |
| T2     | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                        | T2                                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------|----------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

$\beta_{\lambda_5}^{(1)}$

**Quantity:**  $\lambda_5$

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 3

| Column | Representative term structure                                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_5(6\,\text{Tr}\Big(Y_d\,\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,\,(Y_u)^\dagger\Big)-4\lambda_{HS_1}^{(A)})$ |
| T2     | $-\frac{21g_2^2\lambda_5}{2}$                                                                                                                                       |
| T3     | $-\frac{9g_1^2\lambda_5}{2}$                                                                                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                                  | T2                            | T3                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_5(6\,\text{Tr}\Big(Y_d\,\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,\,(Y_u)^\dagger\Big)-4\lambda_{HS_1}^{(A)})$ | $-\frac{21g_2^2\lambda_5}{2}$ | $-\frac{9g_1^2\lambda_5}{2}$ |

Quantity:  $\lambda_5$

Terms 2 of 3

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+2\lambda_{12}^{(1)}\lambda_5$   |
| T2     | $+2\lambda_1\lambda_5$            |
| T3     | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                              | T2                     | T3                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------|------------------------|-----------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

Quantity:  $\lambda_5$

Terms 3 of 3

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |

$$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

**Quantity:**  $\lambda_{HS_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(A)}}{d\mu}=\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

### Terms 1 of 3

| Column | Representative term structure                                                                                                                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{HS_1}^{(A)}\left(6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{(A)}\right)$ |
| T2     | $-\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$                                                                                                                                                                                     |
| T3     | $+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$                                                                                                                                                                       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                  | T2                                              | T3                                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\lambda_{HS_1}^{(A)}\left(6\,\mathrm{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{(A)}\right)$ | $-\frac{8}{3}\lambda_5\left(\lambda_5\right)^*$ | $+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ |

Quantity:  $\lambda_{HS_1}^{(A)}$

Terms 2 of 3

| Column | Representative term structure                |
|--------|----------------------------------------------|
| T1     | $-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$     |
| T2     | $+2\lambda_1\lambda_{HS_1}^{(A)}$            |
| T3     | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                       | T2                                | T3                                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------|-----------------------------------|----------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ |

Quantity:  $\lambda_{HS_1}^{(A)}$

Terms 3 of 3

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Quantity:  $\lambda_{S_1}^{(A)}$

$16\pi^2\mu\frac{d\lambda_{S_1}^{(A)}}{d\mu}=\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Terms 1 of 4

| Column | Representative term structure                                                    |  |
|--------|----------------------------------------------------------------------------------|--|
| T1     | $\frac{3g_2^4}{4}$                                                               |  |
| T2     | $+g_1^2\left(9g_2^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$ |  |
| T3     | $+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$                            |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                 | T2                                                                               | T3                                                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------|----------------------------------------------------------------------------------|-------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3g_2^4}{4}$ | $+g_1^2\left(9g_2^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$ | $+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |

Quantity:  $\lambda_{S_1}^{(A)}$

Terms 2 of 4

| Column | Representative term structure       |
|--------|-------------------------------------|
| T1     | $+ \frac{3 g_1^4}{2}$               |
| T2     | $+ \frac{\lambda_{12}^{(1)2}}{4}$   |
| T3     | $+ \frac{\lambda_{HS_1}^{(1)2}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                    | T2                                | T3                                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------|-----------------------------------|-------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+ \frac{3 g_1^4}{2}$ | $+ \frac{\lambda_{12}^{(1)2}}{4}$ | $+ \frac{\lambda_{HS_1}^{(1)2}}{2}$ |

Quantity:  $\lambda_{S_1}^{(A)}$

Terms 3 of 4

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+3\lambda_{HS_1}^{(A)^2}$                  |
| T2     | $+\frac{7\lambda_{S_1}^{(1)^2}{}}$          |
| T3     | $+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                         | T2                                 | T3                                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------|------------------------------------|---------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)^2}$ | $+\frac{7\lambda_{S_1}^{(1)^2}{}}$ | $+13\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$ |

Quantity:  $\lambda_{S_1}^{(A)}$

Terms 4 of 4

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+22\lambda_{S_1}^{(A)2}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $+22\lambda_{S_1}^{(A)2}$ |

# Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis  $C_{ijab}$ . Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$$\beta_{C_{1133}}^{(1)}$$

**Quantity:** 1, 1, 3, 3

$$16\pi^2\mu\frac{dC_{1133}}{d\mu}=\beta_{C_{1133}}^{(1)}$$

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$ |

$\beta_{C_{1134}}^{(1)}$

Quantity: 1, 1, 3, 4

$$16\pi^2\,_{\mu}\frac{dC_{1134}}{d\mu}=\beta_{C_{1134}}^{(1)}$$

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$ |

$\beta_{C_{1144}}^{(1)}$

Quantity: 1, 1, 4, 4

$16\pi^2\,_{\mu}\frac{dC_{1144}}{d\mu}=\beta_{C_{1144}}^{(1)}$

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                 |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1}y_2}{3MF}$ |

$$16\pi^2\,\mu\frac{dC_{11511}}{d\mu}=\beta_{C_{11511}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$       |
| T2     | $\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                            | T2                                                         | T3                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |

Terms 2 of 2

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$ |
| T2     | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{M F}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                       | T2                                                          |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------------------|-------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3 M F}$ | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12 M F}$ |

$$16\pi^2\,\mu\frac{dC_{11512}}{d\mu}=\beta_{C_{11512}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                  |
|--------|----------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{3^{\frac{3}{4}}i\lambda da12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                             | T2                                                               | T3                                                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |

Terms 2 of 2

| Column | Representative term structure                                          |
|--------|------------------------------------------------------------------------|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1} y_2}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 y \overline{e}}{MF}$   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                       | T2                                                                     |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1} y_2}{3MF}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1} y_2 y \overline{e}}{12MF}$ |

$$16\pi^2\,\mu\frac{dC_{11611}}{d\mu}=\beta_{C_{11611}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,i\,lambda12\,Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$            |
| T3     | $\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                | T2                                                     | T3                                                              |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,i\,lambda12\,Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$ |

Terms 2 of 2

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$              |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                             | T2                                                                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$ |

$$16\pi^2\,\mu\frac{dC_{11612}}{d\mu}=\beta_{C_{11612}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$       |
| T2     | $\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                            | T2                                                         | T3                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |

Terms 2 of 2

| Column | Representative term structure                                        |
|--------|----------------------------------------------------------------------|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                     | T2                                                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------------------------|-----------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$ | $\frac{3^{\frac{3}{4}} y e \overline{y_1 y_2 y e}}{12MF}$ |

$$16\pi^2\,\mu\frac{dC_{11911}}{d\mu}=\beta_{C_{11911}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                  |
|--------|----------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{3^{\frac{3}{4}}i\lambda da12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                            | T2                                                               | T3                                                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |

Quantity: 1, 1, 9, 11

Terms 2 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y \bar{e}}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y \bar{e}}}{12 M F}$ |

$$16\pi^2\,\mu\frac{dC_{11912}}{d\mu}=\beta_{C_{11912}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                           | T2                                                    | T3                                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{11\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{8MF}$ |

Quantity: 1, 1, 9, 12

Terms 2 of 2

| Column | Representative term structure                     |
|--------|---------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$ |

$\beta_{C_{111011}}^{(1)}$

**Quantity:** 1, 1, 10, 11

$$16\pi^2{}_{\mu}\frac{dC_{111011}}{d\mu}=\beta_{C_{111011}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$       |
| T2     | $\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                           | T2                                                         | T3                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |

Quantity: 1, 1, 10, 11

Terms 2 of 2

| Column | Representative term structure                                |
|--------|--------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ |

$\beta_{C_{111012}}^{(1)}$   
Quantity: 1, 1, 10, 12

$$16\pi^2{}_{\mu}\frac{dC_{111012}}{d\mu}=\beta_{C_{111012}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$        |
| T2     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                            | T2                                                           | T3                                                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |

Quantity: 1, 1, 10, 12

Terms 2 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y \bar{e}}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y \bar{e}}}{12 M F}$ |

$\beta_{C_{1213}}^{(1)}$

Quantity: 1, 2, 1, 3

$$16\pi^2\,_{\mu}\frac{dC_{1213}}{d\mu}=\beta_{C_{1213}}^{(1)}$$

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambda T_3 \overline{y_1} y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}lambda T_3 \overline{y_1} y_2}{3MF}$ |

$\beta_{C_{1214}}^{(1)}$

Quantity: 1, 2, 1, 4

$$16\pi^2\,_{\mu}\frac{dC_{1214}}{d\mu}=\beta_{C_{1214}}^{(1)}$$

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$ |

$\beta_{C_{1223}}^{(1)}$

Quantity: 1, 2, 2, 3

$$16\pi^2\,_{\mu}\frac{dC_{1223}}{d\mu}=\beta_{C_{1223}}^{(1)}$$

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$ |

$\beta_{C_{1224}}^{(1)}$

Quantity: 1, 2, 2, 4

$$16\pi^2\,_{\mu}\frac{dC_{1224}}{d\mu}=\beta_{C_{1224}}^{(1)}$$

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                               |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{3}lambdaT_3\overline{y_1}y_2}{3MF}$ |

$$16\pi^2\,\mu\frac{dC_{12711}}{d\mu}=\beta_{C_{12711}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                           | T2                                                    | T3                                                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$ |

Terms 2 of 2

| Column | Representative term structure                                     |
|--------|-------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$      |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$  |
| T3     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                              | T2                                                                | T3                                                                 |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$ |

$\beta_{C_{12712}}^{(1)}$   
Quantity: 1, 2, 7, 12

$$16\pi^2\,\mu\frac{dC_{12712}}{d\mu}=\beta_{C_{12712}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}i\lambda da12Inv1\overline{y_1y_2}}{MF}$   |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                 | T2                                                              | T3                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |

Terms 2 of 2

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$                |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$              |
| T3     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                   | T2                                                             | T3                                                                        |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24 M F}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ |

$\beta_{C_{12811}}^{(1)}$   
Quantity: 1, 2, 8, 11

$$16\pi^2\,\mu\frac{dC_{12811}}{d\mu}=\beta_{C_{12811}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{3^{\frac{3}{4}}\,i\lambda da12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                               | T2                                                                 | T3                                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{25\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\,i\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$ |

Terms 2 of 2

| Column | Representative term structure                                           |
|--------|-------------------------------------------------------------------------|
| T1     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{M F}$   |
| T2     | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$ |
| T3     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                      | T2                                                                         | T3                                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{6 M F}$ | $-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |

$$16\pi^2\,\mu\frac{dC_{12812}}{d\mu}=\beta_{C_{12812}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                           | T2                                                    | T3                                                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$ |

Quantity: 1, 2, 8, 12

Terms 2 of 2

| Column | Representative term structure                                     |
|--------|-------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{MF}$      |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$  |
| T3     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                              | T2                                                                | T3                                                                 |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2y\overline{e}}}{12MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{6MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{6MF}$ |

$\beta_{C_{2211}}^{(1)}$

Quantity: 2, 2, 1, 1

$$16\pi^2\,_{\mu}\frac{dC_{2211}}{d\mu}=\beta_{C_{2211}}^{(1)}$$

| Column | Representative term structure                      |
|--------|----------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambda T_3 \overline{y_1} y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}lambda T_3 \overline{y_1} y_2}{3MF}$ |

$\beta_{C_{2212}}^{(1)}$

Quantity: 2, 2, 1, 2

$$16\pi^2\,_{\mu}\frac{dC_{2212}}{d\mu}=\beta_{C_{2212}}^{(1)}$$

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $\frac{\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                      |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{2\sqrt{3}i\lambda bdaT_3\overline{y_1y_2}}{3MF}$ |

$\beta_{C_{2222}}^{(1)}$

Quantity: 2, 2, 2, 2

$$16\pi^2\,_{\mu}\frac{dC_{2222}}{d\mu}=\beta_{C_{2222}}^{(1)}$$

| Column | Representative term structure                   |
|--------|-------------------------------------------------|
| T1     | $\frac{\sqrt{3}lambdaT_3\overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                 |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{2\sqrt{3}lambdaT_3\overline{y_1}y_2}{3MF}$ |

$$16\pi^2\,\mu\frac{dC_{22511}}{d\mu}=\beta_{C_{22511}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                           | T2                                                    | T3                                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{11\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{8MF}$ |

Quantity: 2, 2, 5, 11

Terms 2 of 2

| Column | Representative term structure                     |
|--------|---------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$ |

$\beta_{C_{22512}}^{(1)}$   
Quantity: 2, 2, 5, 12

$$16\pi^2\,\mu\frac{dC_{22512}}{d\mu}=\beta_{C_{22512}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,i\,lambda12\,Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{MF}$            |
| T3     | $\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                | T2                                                     | T3                                                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,i\,lambda12\,Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1y_2}}{8MF}$ | $\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{8MF}$ |

Quantity: 2, 2, 5, 12

Terms 2 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |

$$16\pi^2\,\mu\frac{dC_{22611}}{d\mu}=\beta_{C_{22611}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                  |
|--------|----------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$            |
| T2     | $\frac{3^{\frac{3}{4}}i\lambda da12Inv1\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{MF}$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                            | T2                                                               | T3                                                    |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{11\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{8MF}$ | $-\frac{3^{\frac{3}{4}}i\lambda da12Inv1\overline{y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}ig_1^2\overline{y_1y_2}}{8MF}$ |

Quantity: 2, 2, 6, 11

Terms 2 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y \bar{e}}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y \bar{e}}}{12 M F}$ |

$$16\pi^2\,\mu\frac{dC_{22612}}{d\mu}=\beta_{C_{22612}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                           | T2                                                    | T3                                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{11\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{8MF}$ |

Quantity: 2, 2, 6, 12

Terms 2 of 2

| Column | Representative term structure                     |
|--------|---------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}}ye\overline{y_1y_2ye}}{12MF}$ |

$$16\pi^2\,\mu\frac{dC_{22911}}{d\mu}=\beta_{C_{22911}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                               | T2                                                                   | T3                                                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

Terms 2 of 2

| Column | Representative term structure                              |
|--------|------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$   |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                          | T2                                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------|---------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2 y e}}{12 M F}$ |

$$16\pi^2\,\mu\frac{dC_{22912}}{d\mu}=\beta_{C_{22912}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$ |
| T2     | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$        |
| T3     | $\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                           | T2                                                    | T3                                                             |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|--------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$ | $\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$ | $\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$ |

Terms 2 of 2

| Column | Representative term structure                                     |
|--------|-------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2ye}{MF}$                 |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                   | T2                                                                  |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|------------------------------------------------------|---------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2ye}{12MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{3MF}$ |

$$16\pi^2{}_{\mu}\frac{dC_{221011}}{d\mu}=\beta_{C_{221011}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$       |
| T2     | $\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                            | T2                                                         | T3                                                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|---------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{3^{\frac{3}{4}}\lambda\overline{Inv1y_1y_2}}{6MF}$ | $-\frac{3^{\frac{3}{4}}g_1^2\overline{y_1y_2}}{8MF}$ |

Terms 2 of 2

| Column | Representative term structure                                     |
|--------|-------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2ye}{MF}$                 |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{MF}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                  | T2                                                                 |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-----------------------------------------------------|--------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $\frac{3^{\frac{3}{4}}ye\overline{y_1}y_2ye}{12MF}$ | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1}y_2}{3MF}$ |

$$16\pi^2\mu\frac{dC_{221012}}{d\mu}=\beta_{C_{221012}}^{(1)}$$

Terms 1 of 2

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{MF}$              |
| T2     | $\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{MF}$ |
| T3     | $\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{MF}$       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                             | T2                                                                 | T3                                                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|----------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{17\cdot 3^{\frac{3}{4}}ig_2^2\overline{y_1y_2}}{24MF}$ | $-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}g_2^2\overline{y_1y_2}}{3MF}$ | $-\frac{3^{\frac{3}{4}}ilambda12Inv1\overline{y_1y_2}}{6MF}$ |

Terms 2 of 2

| Column | Representative term structure                                         |
|--------|-----------------------------------------------------------------------|
| T1     | $\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{M F}$              |
| T2     | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} y \overline{e}}{M F}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$          | T1                                                          | T2                                                                       |
|-------|----------|-----------|-----------|-----------|-----------|-------|----------------|-------------------------------------------------------------|--------------------------------------------------------------------------|
| T3-D  | -2       | 3         | -1        | 1         | 0         | 2     | $-\frac{1}{2}$ | $-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8 M F}$ | $\frac{3^{\frac{3}{4}} i y e \overline{y_1 y_2} y \overline{e}}{12 M F}$ |